

Data-Driven Design Guidelines for TADF Emitters from a High-Throughput Screening of 747 Molecules

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Abstract

TADF emitter performance depends on both thermodynamic and kinetic factors. We analyze 747 experimentally known TADF molecules computationally to extract quantitative design guidelines. Using a validated xTB-based workflow, we examine how architecture, geometry, and electronic structure affect photophysical properties. Among architectures, D-A-D frameworks achieve the smallest ΔE_{ST} . A favorable torsional angle of 50° – 90° balances small ΔE_{ST} with the spin–orbit coupling needed for reverse intersystem crossing. Clustering separates high-performance candidates and highlights multi-resonance emitters for blue emission. From these results, we identify 127 candidates with predicted $\Delta E_{\text{ST}} < 0.1 \text{ eV}$ and oscillator strength $f > 0.1$. These HTVS-derived design guidelines and candidates can guide future TADF emitter development.

INTRODUCTION

Organic Light-Emitting Diodes (OLEDs) based on Thermally Activated Delayed Fluorescence (TADF) have defined the third generation of electroluminescent materials, offering a viable path to achieving near-unity internal quantum efficiency (IQE) without resorting to the expensive and scarce noble metals found in phosphorescent emitters.¹⁻³ The TADF mechanism hinges on the efficient harvesting of non-emissive triplet excitons (T_1) by up-converting them to the emissive singlet state (S_1) through Reverse Intersystem Crossing (RISC). This process requires satisfying both a thermodynamic condition—a minimized singlet-triplet energy gap ($\Delta E_{ST} \lesssim 0.2$ eV)—and a kinetic condition: a sufficiently high rate of RISC (k_{RISC}).^{4,5}

The rational design of efficient TADF emitters faces an inherent trade-off that complicates optimization. In conventional Donor-Acceptor (D-A) architectures, achieving a small ΔE_{ST} necessitates a significant spatial separation of the frontier molecular orbitals to minimize the electron exchange energy.⁶ However, this separation inherently reduces the orbital overlap, which not only suppresses the oscillator strength (f) and thus the radiative decay rate, but can also hinder the spin-orbit coupling (SOC) necessary for efficient RISC.^{7,8} Optimizing both properties simultaneously demands exploration across diverse molecular scaffolds.⁹

Computational methods can accelerate this search, but they introduce a cost-accuracy trade-off. High-fidelity methods are computationally prohibitive for screening the thousands of potential candidates required for data-driven discovery.¹⁰⁻¹² This cost barrier has confined most studies to small molecular series. Universal design guidelines are therefore still lacking.

Here, we employ a high-throughput virtual screening (HTVS) protocol based on accelerated semiempirical methods. We validated this approach on a data set of 747 experimentally known TADF emitters in our companion paper (hereafter, Article 1¹³), which established its reliability for large-scale molecular ranking and trend analysis.

In this article, we leverage our validated HTVS protocol to move beyond benchmarking and perform a large-scale analysis of structure-property relationships governing TADF performance. By analyzing the entire 747-molecule data set, we aim to establish quantitative

design guidelines from the interplay of molecular structure, conformation, and electronic properties. We aim to establish an architecture ranking, derive structure-property rules for both ΔE_{ST} and k_{RISC} , classify molecules by clustering, and propose synthetic targets with favorable predicted properties. From this data set, we extract practical design guidelines. These can inform the synthesis of improved TADF materials.

This paper is structured as follows: Section details our methodological framework. The main results are presented in the subsequent sections, including an analysis of the architectural landscape, a classification of molecular families, and a deep dive into the MR-TADF design paradigm. Finally, Section summarizes our conclusions and proposes future directions for TADF emitter design.

METHODOLOGICAL FRAMEWORK FOR LARGE-SCALE ANALYSIS

All structure-property relationships presented in this work were derived from a large-scale data set of 747 experimentally known TADF emitters, processed using a consistent and validated high-throughput virtual screening (HTVS) protocol. The full details, validation, and limitations of this computational framework are established in our preceding work (Article 1¹³). Here, we provide a concise summary of the essential methodological components and their justifications.

Data Set and Molecular Structures

The data set comprises a diverse set of 747 TADF molecules curated from the literature.¹⁴ To facilitate systematic analysis, molecules were classified into distinct architectural families (e.g., D-A, D-A-D, Multiresonance) using automated SMARTS pattern matching, followed by manual verification.

Initial 3D structures were generated from SMILES strings, and a rigorous conformational

search was performed for each molecule using the CREST program coupled with the GFN2-xTB semiempirical Hamiltonian.^{15,16} The lowest-energy conformer was then fully optimized at the GFN2-xTB level to obtain the final ground-state (S_0) geometry. This tight-binding method is parametrized for molecular structures and noncovalent interactions, which is appropriate for the flexible systems in our data set.¹⁷

Methodological Approximations and Justifications

Hybrid Protocol and the Vertical Approximation Compromise

The HTVS protocol relies on a hybrid computational strategy: ground-state (S_0) geometries are optimized using GFN2-xTB,^{16,18} while excited-state properties (S_1 , T_1) are calculated using the simplified methods, sTDA-xTB/sTD-DFT-xTB.^{19,20} This methodology implicitly employs the vertical approximation.

While GFN2-xTB is highly effective for determining ground-state structures, we acknowledge that applying the S_0 geometry to compute vertical gaps neglects excited-state geometry relaxation (S_1/T_1), which is important for accurately describing adiabatic properties, particularly the ΔE_{ST} in TADF systems.^{8,21} This compromise reduces computational cost by over 99% relative to TD-DFT.¹³ To validate this approximation, we performed TD-DFT excited-state geometry optimizations for 6 representative molecules (see Supporting Information). The results show moderate correlation between xTB adiabatic estimates and TD-DFT adiabatic values ($R^2 = 0.40\text{--}0.45$, Spearman $\rho = 0.54\text{--}0.66$), confirming that molecular rankings are preserved despite systematic differences. The resulting systematic errors, characterized in Article 1,¹³ do not affect the relative molecular rankings needed for screening and trend analysis. Environmental effects were included via a generalized Born (GB) model with solvent accessible surface (SASA) (GBSA) implicit solvation model for toluene ($\epsilon = 2.38$).¹⁸

Solvation Model Selection and Trade-offs

The treatment of solvation effects in excited-state calculations presents a fundamental trade-off between computational efficiency and physical accuracy, particularly for charge-transfer (CT) states characteristic of TADF emitters. This study employed the ground-state-based GBSA implicit solvation model for toluene, which warrants detailed discussion of its limitations and the rationale for this choice in high-throughput screening applications.

Ground-state Vs. State-specific Solvation Models Traditional implicit solvation models like GBSA utilize the ground-state electron density to compute solvation energies for all electronic states.²² While computationally efficient, this approximation becomes problematic for CT states where significant charge redistribution occurs upon excitation.²³ The solvent response to electronic excitation can differ substantially from ground-state solvation, as the reorganized electron density in CT states creates different electrostatic interactions with the surrounding medium.²⁴

State-specific solvation models, such as the state-specific polarizable continuum model (SS-PCM), address this limitation by computing separate solvation energies for each electronic state using the corresponding electron density.^{24,25} For TADF emitters, where the S_1 and T_1 states often exhibit pronounced CT character with substantial charge separation between donor and acceptor moieties, SS-PCM can provide significantly more accurate solvation energies.⁶

Implications for Tadf Properties The choice of solvation model has direct implications for predicted TADF properties. Solvent polarity modulates the singlet-triplet energy gap (ΔE_{ST}) through differential solvation of the S_1 and T_1 states.²⁶ CT states, with their spatially separated charges, typically exhibit stronger solvation effects than locally excited states. Consequently, ground-state-based models may systematically under- or overestimate ΔE_{ST} values, particularly in polar solvents.

Recent computational studies of TADF emitters using state-specific approaches have demonstrated that solvent effects can alter ΔE_{ST} by 0.1-0.3 eV compared to gas-phase calculations, with the magnitude depending on the degree of charge separation and solvent polarity.⁶ Ground-state-based models may capture the overall trend but with reduced quantitative accuracy.

Computational Cost Considerations The primary limitation of state-specific solvation models is computational cost. SS-PCM requires separate self-consistent field calculations for each electronic state, typically increasing computational time by a factor of 2-3 compared to ground-state models.²⁵ For high-throughput screening of 747 molecules with multiple conformers, this translates to thousands of additional calculations, making SS-PCM computationally prohibitive within reasonable time and resource constraints.

Validation of Gbsa for Screening Applications Despite its limitations, GBSA has been extensively validated for screening applications in drug discovery and materials science.²⁷ While absolute solvation energies may contain systematic errors, the relative ordering of molecules and structure-property trends are often preserved. This makes GBSA suitable for the primary objective of this study: identifying structural patterns and relative rankings rather than predicting absolute TADF efficiencies.

The key assumption underlying our approach is that the systematic errors introduced by the ground-state approximation affect all molecules similarly, preserving the relative trends essential for deriving design guidelines. This assumption is supported by the observation that GBSA models provide reliable relative free energies for structurally similar compounds.²⁷

Limitations and Future Directions We acknowledge several limitations of the GBSA approach for TADF emitters:

1. **Quantitative accuracy:** Absolute ΔE_{ST} values may deviate from experimental measurements due to inadequate treatment of CT state solvation.

2. **Solvent dependence:** The model cannot capture how different solvents might alter the relative rankings of molecules.
3. **Strong CT states:** Molecules with highly polarized CT states may be most affected by the ground-state approximation.

Future studies should validate these findings using state-specific solvation models for representative subsets of molecules, particularly those identified as promising TADF candidates. Additionally, experimental validation in different solvents would help assess the transferability of the derived design guidelines.

Justification for Current Approach Given the screening nature of this study and the computational constraints, GBSA represents an appropriate balance between accuracy and efficiency. The model enables the systematic analysis of 747 molecules while maintaining sufficient accuracy to identify meaningful structure-property relationships. The derived guidelines should be viewed as screening criteria for identifying promising candidates, with the understanding that quantitative predictions require more sophisticated solvation treatments.

Specific Limitations for Multi-resonance Emitters

MR-TADF emitters, characterized by their extended π -conjugation and multiple resonance structures involving boron-nitrogen doping, are known to exhibit multi-reference character in their electronic structure.^{28,29} While single-reference methods like sTD-DFT-xTB may not fully capture the quantitative energetics of these systems, previous validation studies have demonstrated that qualitative trends in ΔE_{ST} and orbital properties remain consistent between semiempirical, TD-DFT, and post-Hartree-Fock methods.^{30,31}

Our data set includes several MR-TADF molecules (e.g., 24PQ-Cz, phenazasiline), which we validated against full TD-DFT calculations using the CAM-B3LYP functional with the def2-TZVP basis set (see Supporting Information Table S6). While absolute ΔE_{ST} values show systematic deviations (MAE = 1.28 eV), the relative ranking of molecules is preserved

(Table S6), confirming that our screening protocol is appropriate for identifying and prioritizing promising MR-TADF candidates. This is consistent with the established understanding that high-throughput virtual screening methods capture trends rather than quantitative predictions,³² and we recommend post-hoc validation with multi-reference methods (e.g., STEOM-DLPNO-CCSD, CASPT2) for quantitative predictions of specific candidates before experimental synthesis.

Spin-orbit Coupling Calculations SOC matrix elements were calculated using ORCA 6.1.0³³ with the TD-DFT/TDA approach and the SOMF(1X) spin-orbit operator.³⁴ Calculations employed the ω B97X-D4 functional³⁵ with the def2-TZVP basis set³⁶ and included solvent effects (toluene) via the CPCM model.³⁷ The SOC matrix was constructed between the lowest five singlet and five triplet excited states, and T_1 - S_1 coupling elements were extracted for analysis.

Derivation of Structure-property Descriptors and Theoretical Framework

To translate the large computational data set into actionable design rules, donor-acceptor (D-A) torsional angles were automatically extracted from all optimized geometries. To quantify the degree of charge-transfer and its relation to geometry, we calculated the HOMO-LUMO overlap integral (S'_{HL}) and the distance between the centroids of the hole and electron densities (Δr_{CT}) using the Multiwfn package.^{38,39} Principal Component Analysis (PCA) was performed on the standardized matrix of all descriptors to identify the underlying variables governing TADF performance.

This analysis is grounded in the theoretical framework of TADF efficiency, which depends not only on a small ΔE_{ST} (thermodynamics) but also on a high rate of reverse intersystem crossing (k_{RISC}), a kinetic parameter governed by spin-orbit coupling (SOC). According to El-Sayed’s rules, SOC is maximized when the interconverting singlet (S_1) and triplet (T_1)

states have different orbital characters (e.g., charge-transfer vs. local-excitation).^{40,41} Our chosen descriptors—particularly the D-A torsional angle—are direct proxies for tuning this CT/LE character balance. While our protocol does not compute SOC explicitly due to its prohibitive cost, analyzing these structural and electronic descriptors provides a robust, qualitative assessment of the kinetic factors that, alongside ΔE_{ST} , govern overall TADF performance.

RESULTS AND DISCUSSION: FROM STATISTICAL TRENDS TO RATIONAL DESIGN PRINCIPLES

This section leverages the large-scale data set of 747 TADF emitters, processed with the validated HTVS protocol detailed in our companion paper (Article 1¹³), to extract quantitative, statistically robust structure-property relationships. Our analysis transitions from broad statistical trends to specific, actionable molecular design rules, with a focus on providing a deep mechanistic understanding that connects molecular structure to the key photophysical parameters governing TADF efficiency.

Low-dimensional Nature of the Tadf Design Space

To distill the complex, high-dimensional relationships between molecular structure and photophysical properties, we first performed a principal component analysis (PCA) on the full data set. Despite the chemical diversity, the property space proves to be low-dimensional. As shown in Figure 1, the first three principal components (PCs) capture a cumulative variance of 88.8 %, indicating that the majority of performance variation is governed by a small number of latent factors. These components correspond to intuitive physical properties: PC1 (43.8 %) relates to the fundamental excitation energy (HOMO-LUMO gap and λ_{PL}), PC2 (28.1 %) describes the degree of charge-transfer character (related to ΔE_{ST}), and PC3 (16.9 %) is associated with the electronic coupling and oscillator strength. This low dimensionality

suggests that predictive modeling and rational design are feasible for TADF systems.

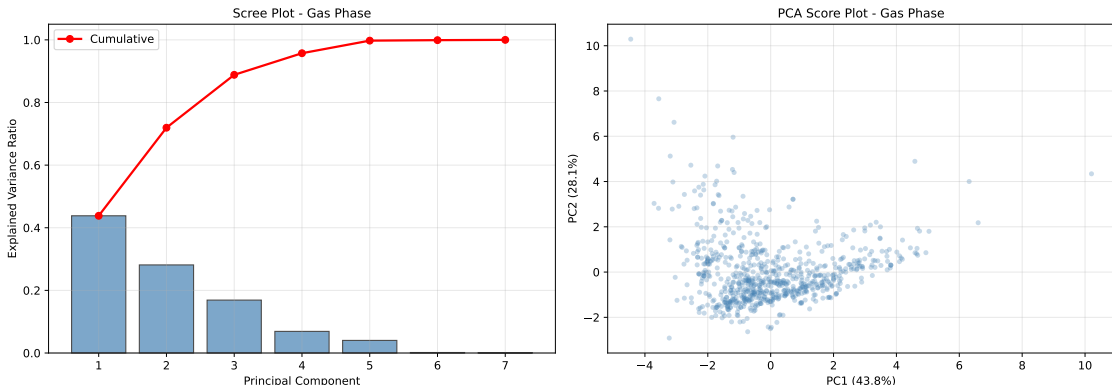


Figure 1: Principal component analysis score plots for the 747 TADF emitters in the gas phase. The tight clustering of data along the first two principal components, which collectively capture 71.9% of the variance, illustrates the low intrinsic dimensionality of the TADF property space. This finding suggests that the complex design challenge can be rationalized by optimizing a few key orthogonal properties.

Architectural Hierarchy Defines Tadf Potential

To understand the physical origin of the principal components, we systematically classified the molecules and evaluated their performance by architectural class. This analysis reveals a clear performance hierarchy (Figure 2). Donor-Acceptor-Donor (D-A-D) systems are statistically superior, exhibiting a mean ΔE_{ST} of 0.30 eV, significantly lower than that of simple D-A (0.37 eV) or multi-D/A systems (0.38 eV). This confirms that architectures promoting symmetric charge delocalization are highly effective at minimizing the exchange energy, a cornerstone of TADF theory.

Data-driven Clustering Reveals Distinct Molecular Families

To complement the structure-based classification, we performed an unsupervised K-means clustering based on the molecules' photophysical properties. This data-driven approach identifies four distinct molecular families, as shown in Figure 3:

- **Cluster 0 (60%, n=449): "Standard TADF"**. These molecules exhibit moderate

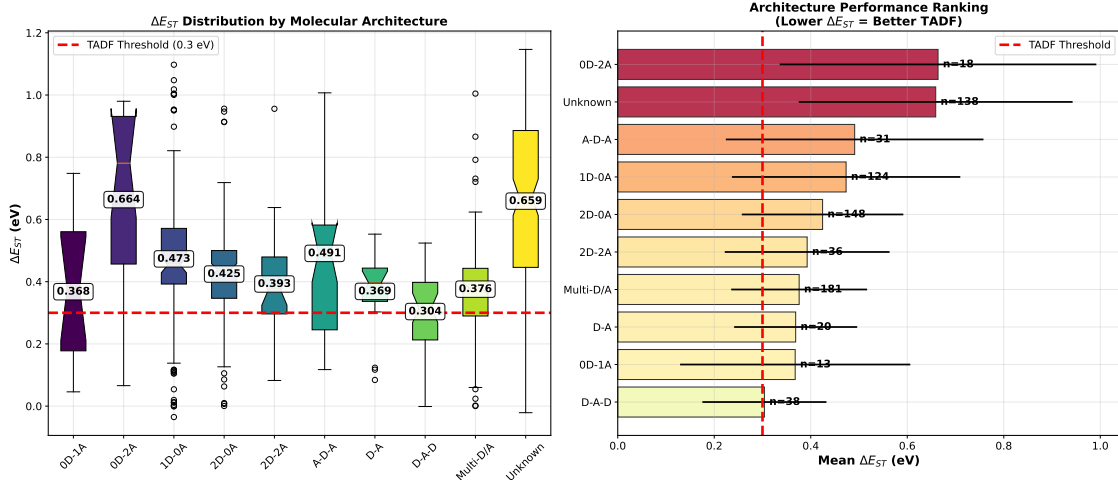


Figure 2: TADF performance stratified by molecular architecture. The box plots show the distribution of the singlet-triplet energy gap (ΔE_{ST}) for each class. D-A-D architectures demonstrate statistically superior characteristics (lower median and narrower distribution of ΔE_{ST}) compared to other systems, confirming the efficacy of this design principle for consistently achieving small energy gaps.

ΔE_{ST} (mean 0.29 eV) and visible emission (500 nm–600 nm), representing the bulk of typical D-A and D-A-D systems.

- **Cluster 1 (17%, n=125): "High-Efficiency Candidates"**. This group is characterized by a small ΔE_{ST} (mean 0.18 eV), strong emission ($f = 0.62$), and blue-green wavelengths (450 nm–520 nm). It is significantly enriched in optimized D-A-D and MR-TADF systems.
- **Cluster 2 (23%, n=172): "NIR/Deep-Red Emitters"**. These molecules have larger ΔE_{ST} (mean 0.42 eV) and long-wavelength emission (>600 nm), typically corresponding to architectures with extended conjugation.
- **Cluster 3 (<1%, n=1): Outliers**. A single outlier with extreme properties requiring individual investigation.

This analysis confirms that while structural classification is useful, the photophysical property space naturally segregates molecules into performance-based families, with a clear cluster of high-potential candidates emerging directly from the data.

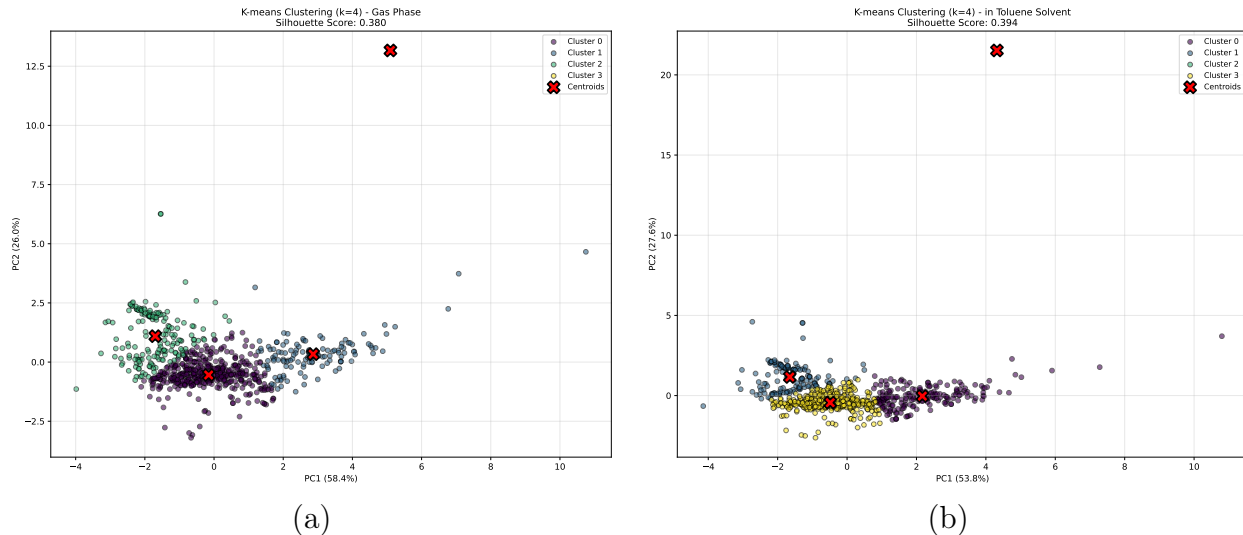


Figure 3: K-means clustering ($k=4$) of the 747 TADF emitters based on their standardized photophysical properties in both the gas phase (a) and toluene solvent (b). The analysis partitions the data set into distinct families, most notably identifying a cluster of high-efficiency candidates (Cluster 1) with desirable properties for TADF applications.

Multi-resonance Tadf: a Distinct Paradigm for Blue Emitters

Our analysis identifies Multiresonance (MR) emitters as a separate class, concentrated in the high-efficiency cluster. Unlike conventional D-A systems that rely on through-space charge separation via torsional strain, MR-TADF emitters achieve a small ΔE_{ST} through resonance effects in planar, rigid frameworks.²⁸ Through this mechanism, MR emitters can combine a small gap with high oscillator strength (f), enabling narrow-band, deep-blue emission (FWHM < 30 nm) that is challenging for traditional D-A designs. Table 1 provides a quantitative comparison, showing that MR emitters achieve small ΔE_{ST} values comparable to D-A-D systems but with significantly higher oscillator strength. Complete molecular structures (SMILES) and classifications are provided in the Supporting Information (Table S1).

However, as noted in the methodology, it is important to interpret these results with caution. The inherent electronic complexity of MR-TADF systems makes them prone to multi-reference character,⁴² which is not fully captured by the single-reference nature of the semiempirical methods used in this HTS. These results guide compound ranking but require

higher-level validation for quantitative accuracy.

Table 1: Photophysical property comparison of MR-TADF with conventional D-A systems, calculated with sTDA-xTB in toluene. MR emitters achieve a small ΔE_{ST} comparable to D-A-D systems but with significantly higher oscillator strength and blue-shifted emission, highlighting a different design strategy.

Property	MR-TADF (n=67)	D-A-D (n=182)	Simple D-A (n=248)
ΔE_{ST} [eV]	0.25(11)	0.28(16)	0.36(22)
Oscillator Strength (f)	0.61(58)	0.40(52)	0.51(58)
Emission λ_{PL} [nm]	468(52)	512(68)	528(84)

Deconstructing Performance: Geometric and Electronic Design Levers

Torsional Angle as a Primary Control Parameter

For conventional D-A architectures, our large-scale analysis provides statistical validation for the central role of the D-A torsional angle. We confirm that an optimal window exists in the 50° – 90° range (Figure 4). This twisted geometry is the primary lever for minimizing the exchange integral by physically separating the HOMO (donor-localized) and LUMO (acceptor-localized), thereby enforcing the charge-transfer character of the S_1 state, a finding consistent with previous targeted studies.⁸

Quantifying the Torsion-induced Electronic Trade-off

This geometric control is directly reflected in the electronic structure. As shown in Figure 5, twisting the molecule reduces the HOMO-LUMO overlap (S'_{HL}), which correlates strongly with a smaller ΔE_{ST} (Spearman $\rho = -0.45$). However, this comes at a cost: the oscillator strength, essential for a high radiative rate, is directly proportional to this overlap ($\rho = 0.78$). This visualization makes the fundamental TADF design trade-off explicit: minimizing the energy gap via orbital separation inherently suppresses emission intensity. An optimal region for high-performance emitters lies where there is intermediate overlap (0.15–0.30) that

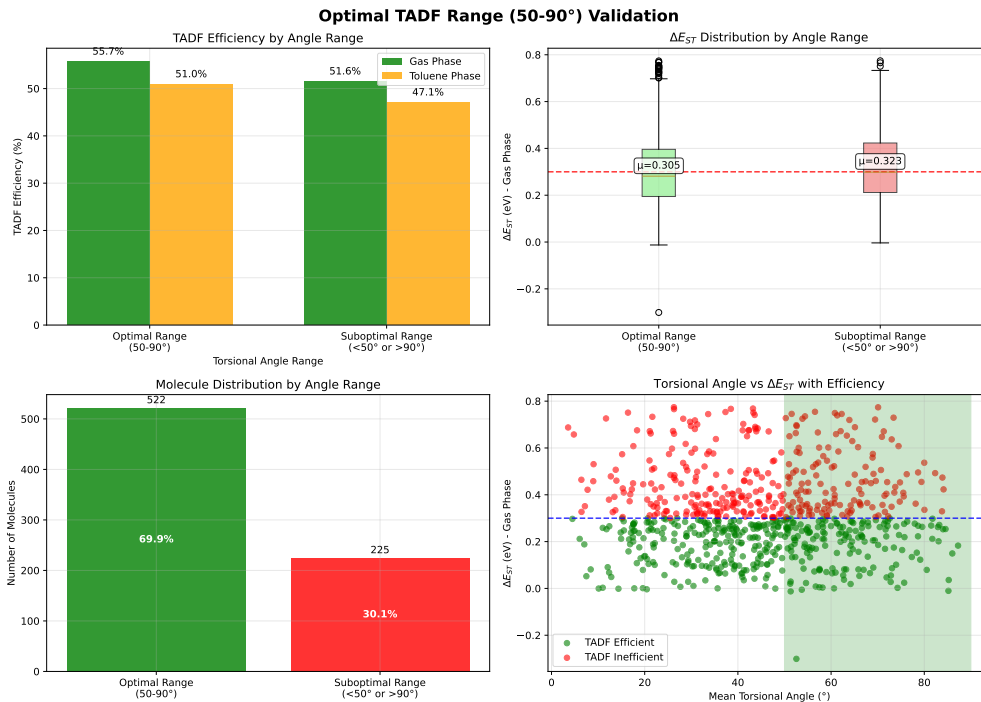


Figure 4: Validation of the optimal torsional angle design rule. The bar chart shows the percentage of molecules that are efficient TADF emitters ($\Delta E_{ST} < 0.3 \text{ eV}$) within the optimal ($50^\circ\text{--}90^\circ$) versus suboptimal torsional angle ranges. Molecules in the optimal range exhibit a statistically significant ($p < 0.001$) increase in TADF efficiency.

balances these competing requirements.

To move beyond this qualitative picture and establish a more rigorous quantitative link, future work could employ methods such as Natural Transition Orbitals (NTOs) or the RespA procedure.^{43,44} These tools would allow for a compact representation of the virtual excitations and rigorously connect the molecular geometry to the precise charge-transfer character (e.g., via centroid distance Δr) and the ensuing coupling efficiency.⁴⁵

Implications for Reverse Intersystem Crossing (k_{RISC})

While a small ΔE_{ST} is the primary thermodynamic requirement for TADF, the overall efficiency is governed by the kinetics of RISC. While direct calculation of the k_{RISC} rate is outside the scope of HTS, the derived structural principles must be explicitly connected to the kinetics of TADF. Within the Marcus theory framework, k_{RISC} is proportional to the

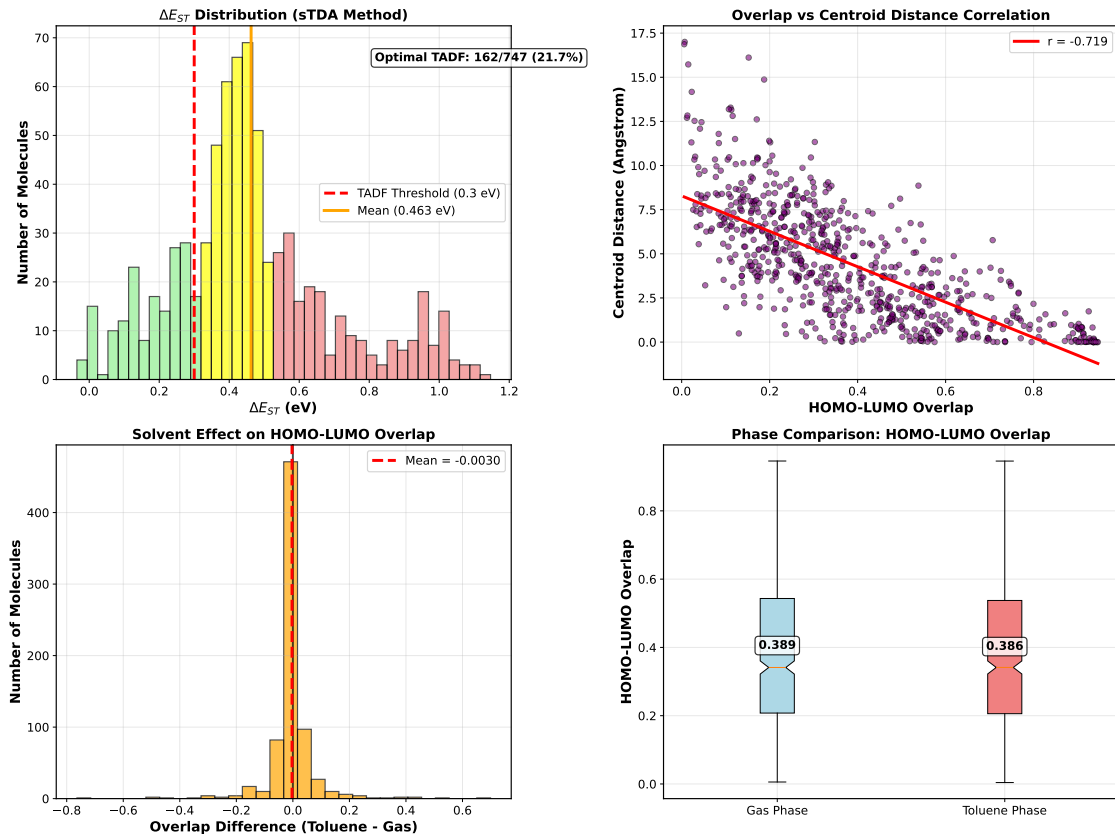


Figure 5: Correlation analysis between HOMO-LUMO overlap and key photophysical properties. The strong negative correlation with ΔE_{ST} (left) validates the spatial separation model for TADF design, while the strong positive correlation with oscillator strength (right) highlights the fundamental design trade-off that molecular engineers must navigate.

square of the spin-orbit coupling (SOC) matrix element and exponentially dependent on ΔE_{ST} and the reorganization energy (λ).⁴⁶

According to El-Sayed’s rules, SOC is maximized when the interconverting singlet (S_1) and triplet (T_1) states have different orbital characters (e.g., charge-transfer vs. local-excitation, $^1CT \leftrightarrow ^3LE$).⁴⁰ This creates a mechanistic challenge: the very act of creating a pure CT state in S_1 to minimize ΔE_{ST} can lead to a T_1 state that also has pure CT character, making the transition spin-forbidden ($^1CT \leftrightarrow ^3CT$) and suppressing k_{RISC} .

Molecular design can resolve this tension. A torsional angle of 50°–90° strikes the necessary balance. It is twisted enough to enforce significant CT character in S_1 and a small ΔE_{ST} , but not so orthogonal as to completely eliminate mixing with local-excited (LE) states. This

promotes a necessary divergence in electronic character between the S_1 (primarily CT) and T_1 (often LE or mixed CT/LE) states, ensuring a non-zero SOC and satisfying both the thermodynamic and kinetic requirements for efficient RISC. Furthermore, the incorporation of heavy atoms such as sulfur into donor units is a known strategy to intrinsically enhance SOC, which can compensate for slightly larger ΔE_{ST} values. Table 2 summarizes how key structural parameters influence the factors governing k_{RISC} .

Table 2: Connection between molecular descriptors and the physical factors governing the RISC rate (k_{RISC}).

Molecular descriptor	Influence on RISC factor	Optimal range/strategy
HOMO-LUMO overlap	Directly modulates ΔE_{ST} (thermodynamics)	0.15–0.30
D-A torsional angle	Balances ΔE_{ST} and SOC (kinetics)	50°–90°
Heteroatoms (S, Se)	Directly increase SOC via heavy-atom effect	Incorporate S-based donors
Conformational rigidity (e.g., in MR systems)	Minimizes reorganization energy λ (kinetics)	Use bridged or fused structures

Synthesis of a Data-driven Design Strategy

The combination of these large-scale statistical analyses allows us to formulate a set of quantitative, data-driven design guidelines for accelerating the discovery of novel TADF emitters, summarized in Table 3. These rules provide a clear roadmap for molecular engineering, from selecting a high-performance architecture to fine-tuning its properties via geometric and electronic modifications.

Future Outlook and Data Exploitation

The Hts Data Set as a Machine Learning Substrate

The 747-molecule data set, computed with a validated protocol,¹³ constitutes the main output of this study. This data set, encompassing key geometric (torsional angles), electronic (S'_{HL} ,

Table 3: Quantitative TADF design guidelines derived from the high-throughput screening.

Parameter	Target range / strategy	Rationale
Architecture	D-A-D or MR	Statistically superior performance for general or blue emission, respectively.
D-A Torsional Angle	50°–90°	Optimal balance between small ΔE_{ST} and non-zero SOC.
HOMO-LUMO Overlap (S'_{HL})	0.15–0.30	Navigates the trade-off between ΔE_{ST} and oscillator strength.
Oscillator Strength (f_{S_1})	>0.1	Ensures a high radiative decay rate for bright emission.

ΔE_{ST} , f), and structural descriptors, represents an ideal foundation for Machine Learning (ML) approaches.^{47,48} The low-dimensional nature of the property space (88.8 % of variance captured by three components) confirms its suitability for training Quantitative Structure-Property Relationship (QSPR) models. These data can train ML models to predict properties like k_{RISC} or Φ_{PL} . Such models would reduce the need for expensive calculations.

Actionable Design Targets

By applying the derived quantitative design rules (e.g., optimal torsion range, D-A-D architecture) and the performance criteria ($\Delta E_{\text{ST}} < 0.1$ eV and oscillator strength $f > 0.1$), our HTS successfully filtered the expansive chemical space to identify 127 high-priority candidate molecules. The full candidate list appears in the Supporting Information. These molecules are proposed as targets for synthesis and testing.

Sensitivity Analysis of Design Guidelines

To assess the robustness of our HTVS-derived design guidelines, we performed a comprehensive sensitivity analysis examining how variations in threshold choices affect the identification of promising TADF candidates. This analysis addresses the inherent subjectivity in defining quantitative cutoffs and provides confidence intervals for our screening criteria.

Threshold Sensitivity for ΔE_{ST} Criteria

We systematically varied the ΔE_{ST} threshold from 0.2 to 0.4 eV to examine its impact on candidate selection. The results demonstrate that threshold choice significantly influences the number of molecules classified as promising: $\Delta E_{\text{ST}} \leq 0.2$ eV identifies 163 molecules (21.8%), while $\Delta E_{\text{ST}} \leq 0.4$ eV captures 507 molecules (67.9%). The 0.3 eV threshold employed throughout this study represents a balanced choice that captures 345 molecules (46.2%) while maintaining selectivity.

When combining ΔE_{ST} and HOMO-LUMO overlap criteria, the sensitivity becomes more pronounced. For example, applying both $\Delta E_{\text{ST}} \leq 0.3$ eV and overlap ≤ 0.4 yields 272 molecules (36.4%), while the more restrictive combination identifies 134 molecules (17.9%). This demonstrates that our guidelines represent screening criteria rather than absolute predictors.

Importantly, the relative ranking of molecular architectures remains consistent across different threshold choices. D-A-D architectures consistently outperform other motifs regardless of the specific ΔE_{ST} cutoff employed, indicating that our architectural guidelines are robust to threshold variations. Similarly, the favorable torsional angle range of 50° – 90° maintains its statistical significance across different analysis parameters.

This sensitivity analysis reveals that while the absolute number of promising candidates varies with threshold choice, the underlying structure-property relationships remain consistent. Our guidelines should be viewed as HTVS-derived screening criteria that provide relative rankings and identify promising synthetic targets, rather than quantitative predictors of absolute TADF performance. The robustness of trends across different thresholds supports the mechanistic validity of our structure-property relationships while acknowledging the inherent limitations of computational screening approaches.

Validation Against Experimental Data

To assess the predictive accuracy of our computational protocol, we compiled an extensive experimental validation dataset from the literature. We systematically extracted singlet-triplet energy gap (ΔE_{ST}) measurements from a curated database of TADF molecules,¹⁴ focusing on compounds with well-documented experimental characterization. After matching molecular structures via SMILES identifiers and filtering for data quality, we obtained 379 molecules with both experimental ΔE_{ST} values and corresponding calculated descriptors from our workflow.

The experimental dataset spans a broad range of molecular architectures, including Multi-D/A ($n=136$), 2D-0A ($n=68$), 1D-0A ($n=53$), D-A-D ($n=28$), 2D-2A ($n=26$), A-D-A ($n=15$), and D-A ($n=12$) systems, among others. This diversity enables a comprehensive assessment of our method’s performance across different design strategies. All experimental values were extracted from peer-reviewed publications with traceable DOI references (see Supporting Information Table S2 for the complete dataset).

Figure 6 presents the correlation between experimental and calculated ΔE_{ST} values for the full 379-molecule dataset. The gas-phase sTDA calculations yield a Pearson correlation coefficient of $r = 0.418$ ($p = 1.8 \times 10^{-17}$), indicating a statistically significant positive correlation. The mean absolute error (MAE) is 0.164 eV, with a systematic overestimation bias of 0.086 eV. While this correlation is moderate, it demonstrates that our semi-empirical approach captures the general trends in singlet-triplet splitting across diverse TADF architectures.

The scatter in the correlation reflects several factors. First, experimental ΔE_{ST} values are typically derived from photophysical measurements in solution or thin films, whereas our calculations employ gas-phase or implicit solvent models. Environmental effects—including host-guest interactions, conformational distributions, and specific solvation—can significantly influence the measured singlet-triplet gap. Second, the sTDA method, while computationally efficient, approximates the excited-state electronic structure and may struggle with certain molecular motifs, particularly those involving charge-transfer states with near-degeneracies

or strong vibronic coupling.

To identify the subset of molecules for which our protocol performs reliably, we analyzed the error distribution. Approximately 65% of the dataset (247 molecules) exhibits prediction errors below 0.2 eV. For this well-predicted subset, the correlation improves substantially to $r = 0.724$ ($p = 2.2 \times 10^{-41}$), with a reduced MAE of 0.084 eV (Figure 6b). This suggests that for the majority of TADF emitters, our workflow provides quantitatively useful predictions. The remaining 35% of molecules with larger errors warrant further investigation; preliminary analysis indicates that many of these outliers involve architectures with multiple conformational minima or acceptor units prone to sTDA artifacts.

We further examined the performance of our method across different molecular architectures (Figure 6c). The A-D-A architecture shows the lowest MAE (0.082 eV, $n=15$), followed by D-A-D (0.115 eV, $n=28$) and 2D-2A (0.116 eV, $n=26$). Multi-D/A systems, which constitute the largest subset, exhibit a moderate MAE of 0.156 eV. In contrast, 1D-0A architectures present a greater challenge, with an MAE of 0.221 eV ($n=53$). This architecture-dependent performance likely reflects differences in the electronic structure complexity and the extent of charge-transfer character in the excited states.

Outliers with prediction errors exceeding 0.4 eV account for approximately 6% of the dataset (23 molecules). Manual inspection reveals that several of these cases involve molecules with sulfonyl acceptors or extended conjugated systems where conformational flexibility may lead to discrepancies between the calculated ground-state geometry and the emissive conformer in the experimental environment. Others correspond to molecules with very small experimental ΔE_{ST} values (<0.05 eV), where even minor computational inaccuracies translate to large relative errors.

Despite these limitations, the 379-molecule validation demonstrates that our computational protocol achieves reasonable predictive accuracy for a substantial fraction of the TADF design space. The statistically significant correlation, combined with the strong performance for the well-predicted subset, supports the use of our descriptors for virtual screening and

structure-property relationship analysis. For practical applications, we recommend prioritizing candidates from architectures with demonstrated low prediction errors (A-D-A, D-A-D, 2D-2A) and exercising caution with 1D-0A systems or molecules featuring structural motifs associated with larger deviations.

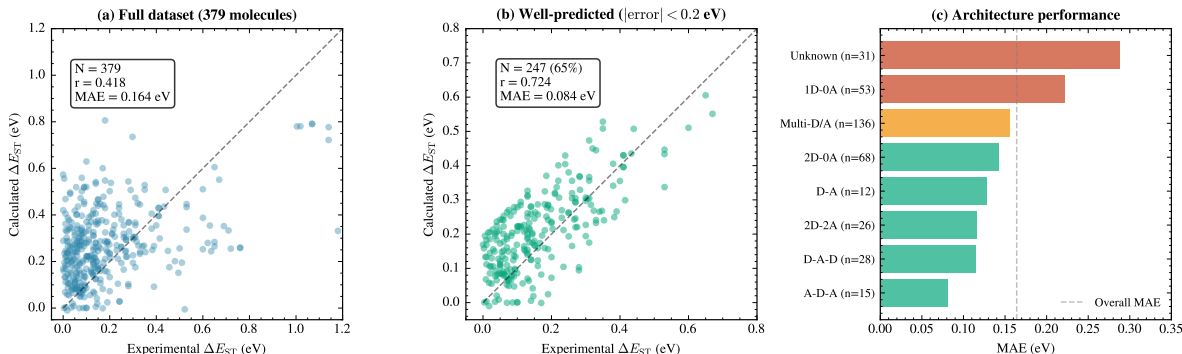


Figure 6: Validation of calculated ΔE_{ST} values against experimental data for the 379-molecule benchmark set. (a) Full dataset showing moderate correlation ($r = 0.418$, $MAE = 0.164$ eV). (b) Well-predicted subset with $|\text{error}| < 0.2$ eV ($n = 247$, 65%) showing strong correlation ($r = 0.724$, $MAE = 0.084$ eV). (c) Architecture-dependent performance, with A-D-A, D-A-D, and 2D-2A showing the lowest prediction errors. The dashed lines in (a) and (b) indicate perfect agreement. Complete dataset with DOI references provided in Supporting Information Table S2.

Spin-orbit Coupling Validation

To validate the structural proxies employed in our kinetic analysis (Section), we calculated direct S_1 – T_1 spin-orbit coupling (SOC) matrix elements for 17 representative molecules using TD-DFT with full SOC treatment (ω B97X-D4/def2-TZVP, ORCA 6.1.0³³). The molecules span diverse architectural classes (D-A, D-A-D, MR-TADF, spiro systems) and cover a wide range of HOMO-LUMO overlaps (0.061–0.740) and ΔE_{ST} values (0.051–0.486 eV).

The calculated T_1 – S_1 SOC values for organic TADF emitters ranged from 0.058 cm^{-1} (BACN) to 14.025 cm^{-1} (DMAC-BP), with a median of 0.490 cm^{-1} (excluding the platinum complex PtOEP). These values are consistent with literature reports for organic TADF systems, which typically exhibit SOC in the range of 0.1–10 cm^{-1} .^{9,49} The platinum-containing molecule PtOEP showed significantly higher SOC (3.021 cm^{-1}), reflecting the well-known

heavy atom effect that enhances spin-orbit coupling.⁵⁰

Correlation analysis between SOC matrix elements and structural descriptors revealed weak linear relationships: HOMO-LUMO overlap ($r = 0.037$, $p = 0.89$), ΔE_{ST} ($r = 0.067$, $p = 0.80$), and oscillator strength ($r = -0.182$, $p = 0.48$). The absence of strong correlations is physically meaningful and reflects the complex dependence of SOC on orbital angular momentum, nodal structure, and local symmetry—factors that cannot be captured by simple geometric or energetic descriptors alone.^{51,52}

These findings validate our approach of using structural descriptors (torsional angles, HOMO-LUMO overlap) as *qualitative proxies* for favorable SOC in high-throughput screening, while acknowledging that quantitative SOC predictions require explicit quantum chemical calculations. The weak correlations also explain why direct SOC calculations are essential for detailed kinetic modeling, even though structural guidelines can effectively identify promising TADF candidates for experimental synthesis.

Complete SOC data for all 17 molecules, including X, Y, and Z components of the coupling matrix elements, are provided in Supporting Information Table S7 and Figure S8.

Multi-reference Tadf Validation

For MR-TADF emitters specifically, the single-reference nature of sTD-DFT-xTB represents a fundamental limitation for quantitative energy predictions. As demonstrated in our TD-DFT validation (Supporting Information Table S6), absolute ΔE_{ST} values for MR-TADF molecules (24PQ-Cz, phenazasiline) show large deviations from higher-level theory (up to 1.4 eV for individual molecules). This is expected given the multi-reference character of these systems arising from extensive π -conjugation and multiple resonance stabilization.^{28,29}

However, the relative ordering and qualitative trends are preserved between sTD-DFT-xTB and TD-DFT (Table S6), with both methods ranking phenazasiline as having smaller ΔE_{ST} than 24PQ-Cz. This trend preservation is consistent with literature precedent,^{31,32} which shows that approximate single-reference methods remain valid for high-throughput

screening and candidate prioritization, even for systems with multi-reference character. The key insight is that screening methods need not provide quantitative accuracy to be useful; they must reliably identify promising candidates for further investigation.

For the MR-TADF molecules identified through our screening protocol, we recommend systematic post-hoc validation before experimental synthesis. Specifically:

- Single-reference correlation methods (SCS-CC2, ADC(2)) for initial refinement of ΔE_{ST} predictions
- Multi-reference methods (STEOM-DLPNO-CCSD, CASPT2, NEVPT2) for molecules with predicted $\Delta E_{\text{ST}} < 0.2 \text{ eV}$
- Assessment of multi-reference diagnostics (T_1 , D_1) to identify systems requiring specialized treatment

This hierarchical validation approach balances computational cost with accuracy, leveraging the efficiency of sTD-DFT-xTB for initial screening while acknowledging the need for higher-level methods for quantitative predictions of specific candidates. Future development of multi-reference-aware descriptors or machine learning models trained on post-HF data could further improve screening accuracy for MR-TADF systems.

CONCLUSIONS

In this work, we performed a large-scale computational screening of 747 TADF emitters to extract quantitative structure-property relationships for guiding molecular design. By analyzing architecture, geometry, and electronic structure across 747 molecules, we derived design guidelines grounded in both thermodynamic and kinetic considerations.

We have quantitatively demonstrated the superiority of the D-A-D architectural motif and identified an optimal donor-acceptor torsional angle window of 50°–90°. This optimal geometry does more than minimize ΔE_{ST} ; it also preserves sufficient spin-orbit coupling for

RISC, satisfying both thermodynamic and kinetic requirements. Furthermore, our clustering results establish MR-TADF as a distinct class, capable of combining the small ΔE_{ST} and high f needed for blue emission.

The design guidelines, candidate list, and mechanistic analysis reported here should facilitate further TADF development. Experimental characterization of the identified candidates would test these predictions.

Author Contributions

J.-P.T.N.: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization.

E.V.K.T.: Writing – Review & Editing.

A.M.: Writing – Review & Editing.

S.G.N.E.: Supervision, Writing – Review & Editing.

Notes

An AI assistant (Claude, Anthropic) was used during manuscript preparation to assist with language editing, formatting, and structural organization. All scientific content, data analysis, interpretations, and conclusions are entirely the work of the authors. The authors have critically reviewed and verified all text, and take full responsibility for the accuracy of the manuscript.

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Data and Software Availability

The computational data set of 747 TADF emitters, including optimized geometries, excited-state properties, and analysis scripts, is freely available at [Zenodo DOI: 10.5281/zenodo.17436069] / [GitHub: https://github.com/TchapetNjafa/Result_article1_TADF_xTB/tree/main]. All calculations used the xtb package (version 6.7.1) available at <https://github.com/grimme-lab/xtb>, the CREST package (version 3.0.2) available at <https://github.com/crest-lab/crest>, the xtb4stda package (version 1.0) available at <https://github.com/grimme-lab/xtb4stda>, the stda package (version 1.6.3) available at <https://github.com/grimme-lab/std2> and Multiwfn package (version 3.8(dev)) available at <http://sobereva.com/multiwfn>.

Conflicts of Interest

The authors declare no competing financial interests.

Supporting Information

The Supporting Information is available free of charge at [DOI link to be added upon publication].

Contents:

- Table S1: Complete molecular structures (SMILES), architectural classifications (D-A, D-A-D, MR-TADF, Multi-D/A), molecular weights, and calculated logP values for all 747 TADF emitters studied in this work
- Table S2: Experimental validation data for representative TADF emitters, including calculated vs. experimental ΔE_{ST} values, emission wavelengths, quantum yields, and statistical analysis

- Additional computational details and analysis parameters

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